

18.Perceptron-Based IRIS Classification

Aim

To implement **IRIS flower classification using the Perceptron algorithm** in Python and classify iris flowers into their respective species.

Algorithm

1. Import the required libraries.
2. Load the Iris dataset.
3. Select the input features and target classes.
4. Split the dataset into training and testing data.
5. Create a Perceptron classifier.
6. Train the model using the training data.
7. Predict the classes of the test data.
8. Calculate the accuracy of the model.
9. Predict the class of a new iris flower.

Program:

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.linear_model import Perceptron

from sklearn.metrics import accuracy_score


iris = load_iris()


X = iris.data

y = iris.target


X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)
```

```
model = Perceptron(max_iter=1000, random_state=42)
```

```
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("Actual Classes:", y_test)
```

```
print("Predicted Classes:", y_pred)
```

```
print("Accuracy:", accuracy * 100, "%")
```

```
new_flower = [[5.1, 3.5, 1.4, 0.2]]
```

```
prediction = model.predict(new_flower)
```

```
print("New Flower:", new_flower)
```

```
print("Predicted Flower:", iris.target_names[prediction[0]])
```

Output:

```
y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Actual Classes:", y_test)
print("Predicted Classes:", y_pred)
print("Accuracy:", accuracy * 100, "%")

new_flower = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(new_flower)

print("New Flower:", new_flower)
print("Predicted Flower:", iris.target_names[prediction[0]])
```

```
... Actual Classes: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0]
Predicted Classes: [1 0 1 1 1 0 1 1 1 1 1 0 0 0 0 1 1 1 1 1 0 1 0 1 1 1 1 1 0 0]
Accuracy: 63.33333333333333 %
New Flower: [[5.1, 3.5, 1.4, 0.2]]
Predicted Flower: setosa
```

Result Statement

Thus, the **Perceptron-based IRIS classification** was successfully implemented in Python, and the species of a new iris flower was successfully predicted.